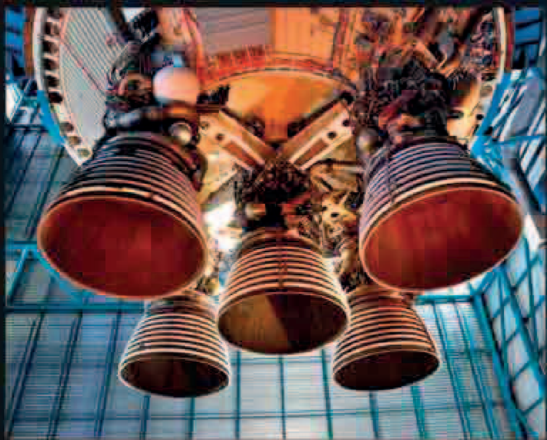
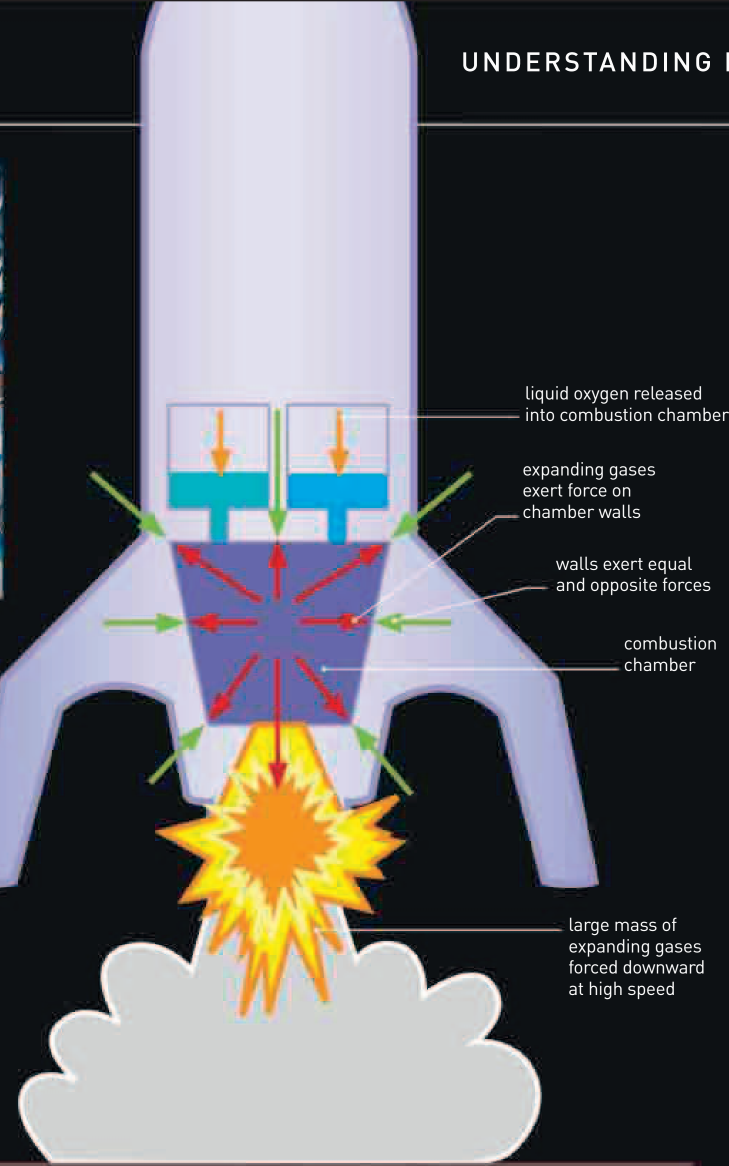


ROCKET ENGINE NOZZLES
Causing a rocket to accelerate upward requires enormous forces, not least to overcome the gravity pulling the rocket downward. The force is generated inside the engine by expanding exhaust gases, which escape through these nozzles.



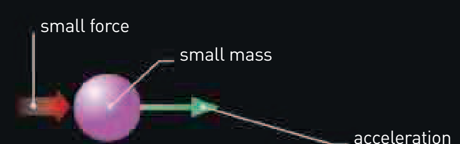
LIFT OFF
Hot gases expand, exerting forces on the walls of the combustion chamber to lift the rocket. The walls of the chamber produce a reaction force that pushes back on the gases, which escape at high speed through the bottom of the engine.

SATURN V
The Saturn V rocket, used during NASA's Apollo missions of the 1960s and 1970s, had a weight of 28 million newtons, and an engine thrust of 34 million newtons.

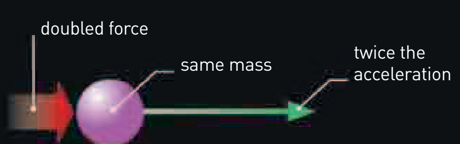
SECOND LAW

Newton's second law involves momentum: an object's mass multiplied by its velocity. The law states that the change in momentum is proportional to the force exerted. So, a force doubled will accelerate an object twice as much; but the same force applied to an object with twice the mass will produce only half the acceleration. The second law is often summarized with a simple equation: $a = F/m$, in which a is the acceleration, F is the force, and m is the mass of the object.

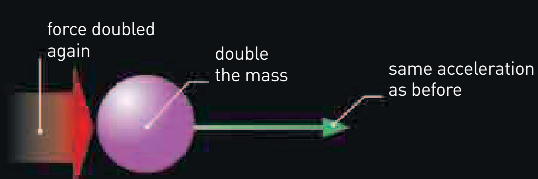
SMALL MASS, SMALL FORCE
An applied force causes an object to accelerate. The acceleration—change in velocity per second—depends upon the size of the force, but also on the mass of the object.



SMALL MASS, DOUBLE FORCE
Since $a = F/m$, doubling the force but keeping the same mass will cause the object to accelerate at twice the rate.

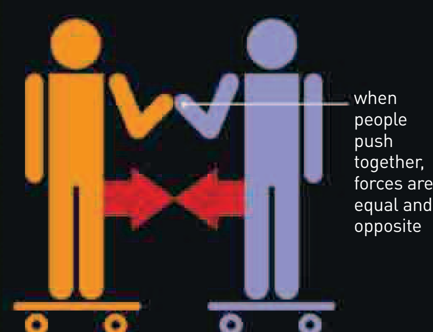


DOUBLE MASS, DOUBLE FORCE
Doubling the force again (to four times the original value) but also doubling the mass produces the same acceleration as before.

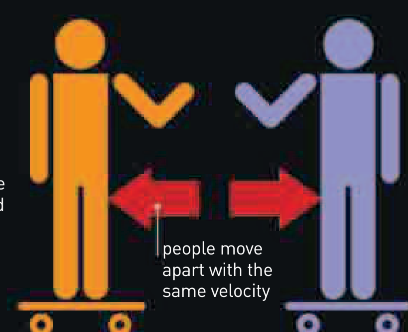


THIRD LAW

Newton's third law states that forces exist in pairs. When one object exerts a force on another, the second object exerts an equal and opposite force on the first. If one of the objects is immobile, then the other object will move; push against a wall on a ice rink, and the wall pushes back on you—which makes you slide on the ice. If both objects can move, then the object with less mass will accelerate more than the other; for example, a heavy gun recoils slightly as the bullet shoots out at high speed.



EQUAL AND OPPOSITE
Two people on skateboards pushing against one other will move apart. Even if only one person does the pushing, the other person's body will produce a reaction force of equal strength in the opposite direction.



EQUAL MASSES
If the two people have the same mass, they will accelerate equally; but if one has much less mass, he or she will move away more quickly, since the same force will produce a greater acceleration.